GlucoTwin | System Architecture

Retrospective CGM forecasting • uncertainty estimation • held-out evaluation

RESEARCH PROTOTYPE

01 / CORE FORECASTING PIPELINE

DATA INPUT

Historical CGM records

95,619 model-ready rows
80 CGM identities · 33 features

PREPARE HISTORY

Causal state checks

Select patient + timestamp
96-step history eligibility
15-minute sampling cadence

PREDICTION ENGINE

Saved horizon models

LightGBM model artifacts
+30 / +60 / +90 / +120 min
No dashboard-time retraining

USER INTERFACE

Streamlit dashboard

Historical replay + forecasts
Intervals, tables and metrics
Interactive research workflow

02 / DASHBOARD OUTPUTS

FORECAST VALUES

Four future horizons

+30 · +60 · +90 · +120 minutes
Predicted glucose values in mg/dL

UNCERTAINTY

Conformal intervals

80% and 90% prediction intervals
Use saved calibration radii

HELD-OUT EVALUATION

Retrospective performance

MAE · RMSE · R^2 · interval coverage
Reported from saved test outputs

03 / SEPARATE SYNTHETIC EHR DEMONSTRATION

SYNTHETIC SOURCE

Synthea-generated EHR CSV records

SYNTHETIC EHR SUMMARY

Demographics + condition, observation and medication counts

DASHBOARD SIDE PANEL

Display-only demo
separate from forecast path

BOUNDARIES / RESPONSIBLE USE

Not model input · linkage not verified
No live CGM · not clinically validated

RECORDED SPLIT · Train 67,532 / Validation 10,713 / Test 17,374

Research and education only — retrospective results are not clinical validation or treatment guidance.